Deal or No Deal AI

Project 3B — Multi-Agent Deal Review Pipeline

Weekly submission report — Draft
Reporting date: October 25, 2026, Pacific Time
Tagline: Every deal has a story. Our agents find the plot holes.
Scope: Local educational prototype using fictional commercial-loan documents.

1. Executive summary

Deal or No Deal AI helps a reviewer answer four questions: What did the documents say? What do the numbers show? Which requirements remain unresolved? What should a human reviewer do next?

The Streamlit application coordinates specialized extraction, financial, compliance, risk and synthesis responsibilities through LangGraph. Live extraction and synthesis use Nebius; financial arithmetic, configured policy checks, risk aggregation, readiness routing and memo assembly use Python. This is a hybrid multi-agent workflow, not a claim that every specialist is an autonomous LLM.

An uploaded Standard Deal has completed actual live generation and a browser walkthrough, including source inspection and memo download. A newer general-upload path supports reviewer-declared reporting context, parallel specialist branches and mandatory evidence checkpoints; its broader behavior is established by offline integration tests, not a completed live walkthrough of every possible document.

The key demonstration is a useful disagreement: a borrower can have satisfactory annual ratios and an illustrative Low risk score while still being **Pending Review** because covenant and evidence checks are incomplete. The system preserves the original AI proposal, deterministic readiness and human decision separately.

2. Assignment alignment

Assignment requirement	Implemented responsibility	Evidence and boundary
Agent 1 extracts key terms	Extraction Agent	Live structured Nebius output with source handles; saved examples remain separately labeled
Agent 2 checks compliance	Compliance Agent	Deterministic configured sample-policy checks; no claim of regulatory certification
Agent 3 flags risks and summarizes	Risk and Summary Agent role	Python risk stage followed by live synthesis; distinct internal nodes
Orchestrator coordinates the pipeline	LangGraph Orchestrator Agent	State, dependencies, conditional routing, checkpoints and explicit continuation
Three or more specialist roles	Extraction, Compliance, Risk/Summary; financial support	Two normal LLM stages, not three independent LLM agents
Agent communication	Structured shared-state artifacts	Source-linked panel outputs and recorded handoffs; no fabricated agent chat

Assignment requirement	Implemented responsibility	Evidence and boundary
Sample review output	Source-linked Markdown credit memo	Actual completed uploaded Standard Deal evidence retained
Recording and repository links	Submission artifacts	**[NOT VERIFIED]** Links still need to be supplied

3. Business problem and opportunity

A commercial-loan review combines contracts, financial schedules, collateral information and policy requirements. A reviewer must reconcile reporting periods, calculate ratios, identify missing evidence and prepare a defensible explanation. Fragmented documents make those tasks repetitive and make inconsistencies harder to notice.

A single free-form model response can mix extraction, arithmetic and judgment without exposing which step failed. This prototype separates those responsibilities and makes intermediate results visible before a human records a decision.

The intended business opportunity is less document transcription and memo assembly, more consistent application of explicit rules, and earlier identification of unresolved issues. These benefits are proposed; analyst time savings and lending outcomes have not been measured.

4. Product vision and target users

User	Job to be done	Product support
Credit analyst / loan officer	Prepare a coherent deal review	Upload inspection, extracted terms, calculations and draft memo
Underwriter / Risk Manager	Challenge repayment assumptions and missing inputs	Source evidence, ratio definitions, risk drivers and evidence-review controls
Compliance Officer	Review applicable controls and policy findings	Explicit applicability, exceptions and policy-linked evidence
Credit officer / committee member	Understand the recommendation and decide what happens next	Readiness, unresolved issues, comments and role-controlled human actions

Roles are simulated. The application assists decisions; it does not originate, fund or authorize a real loan. Insurance and separate equity-investment reviews are possible future applications, not implemented MVP workflows. The investment perspective in this MVP concerns the same loan as a debt investment.

5. Product features and user experience

Screen	User objective and capabilities	Backend involvement and output
Deal Dashboard	Find deals needing attention	Scoped stored review states, readiness and human status; navigable deal summaries

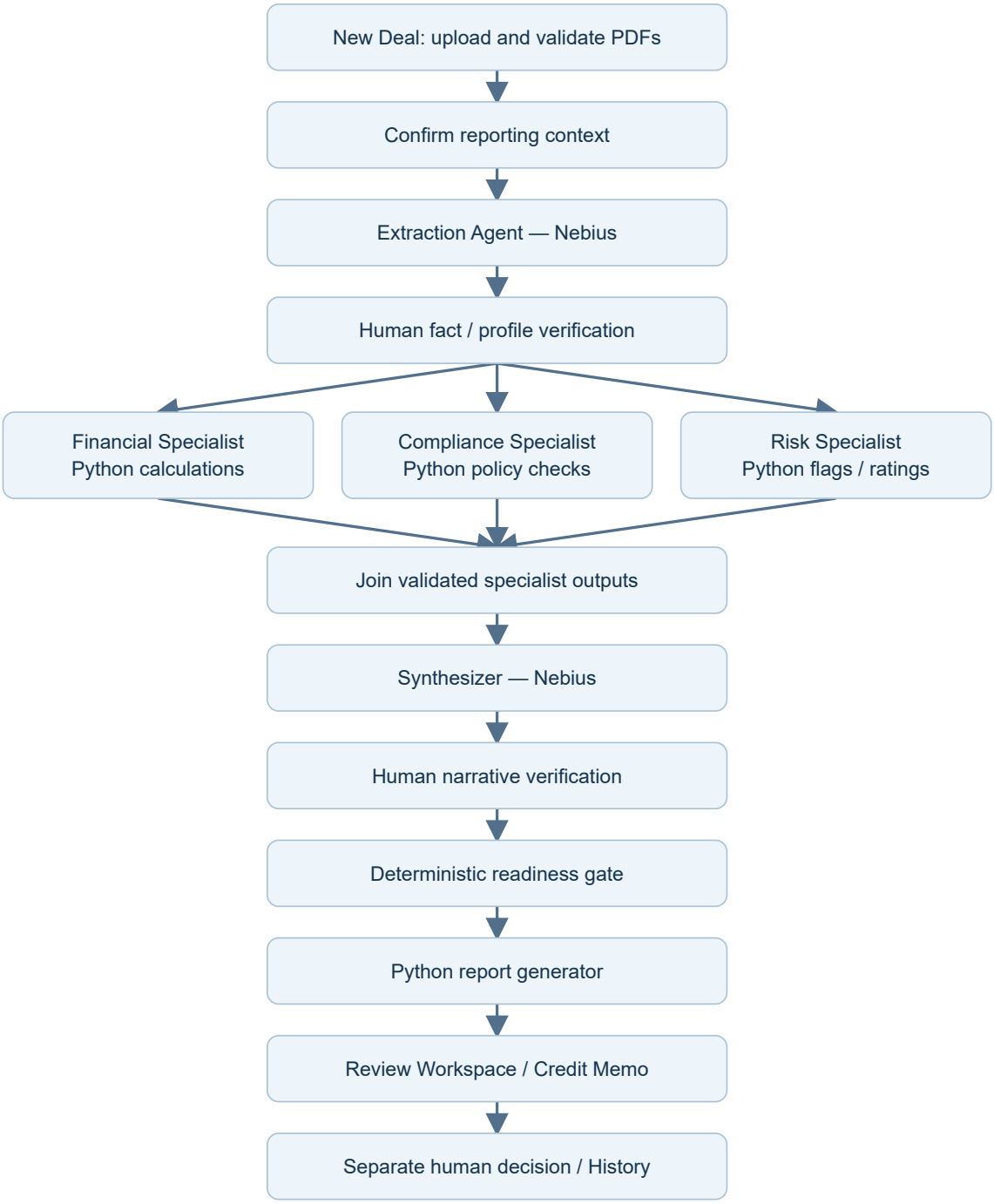
Screen	User objective and capabilities	Backend involvement and output
New Deal	Start a sample or upload one fictional deal package	Parse/validate PDFs, show page evidence, prefill editable reporting context and explicitly start analysis
Review Workspace	Understand and challenge findings	Extracted terms, financial metrics, policy results, risks, evidence explorer and human-review controls
Agent Activity	See what executed and where work paused	Recorded stage events, specialist outputs, handoffs and failures; no invented execution history
Credit Memo	Read and export a consolidated review	Python-generated Markdown with findings, conditions, limitations and supporting references
Decision History	Review who did what and why	Separately recorded comments, status changes, timestamps and analysis snapshots

The reporting-context form uses document-derived suggestions where text is clear. It retains editable dates, annual actual/projection type and policy applicability; comparative dates are optional. Suggestions are not verification. Unrecognized or conflicting fields remain blank; policy applicability remains Unknown unless explicitly declared. Review date may default to the upload's UTC date, visibly identified as a default rather than a contractual date.

All downstream screens use the same scoped Deal ID, review identity, source binding and stored artifacts. Saving reporting metadata does not execute AI.

6. Multi-agent architecture diagram

The diagram below describes the **new uploaded-review path**. Registered sample and historical reviews retain their recorded sequential or demo-draft behavior.



LangGraph coordinates state, checkpoints and stage routing; SQLite retains records. Agent Activity reads recorded events. Diagram: new uploaded-review path.

7. Agent responsibilities and orchestration

Component	Receives	Produces and validates	Handoff
Extraction Agent	Current deal's parsed pages, source handles, reporting bases and strict schema	DealTermsWire: terms, financial inputs, evidence references and conflicts; Pydantic and source/basis checks	Human fact/profile verification

Component	Receives	Produces and validates	Handoff
Financial Specialist	Extracted values with units, periods and scope	Reproducible Decimal calculations; missing or invalid inputs remain explicit	Joined financial panel
Compliance Agent	Extracted facts, independently derived metrics and configured sample policy	Applicability-aware rule results, missing controls and exceptions	Joined compliance panel
Risk Agent	Current deal's facts, derived metrics and policy findings	Risk flags; weighted score only when supported authored factors exist	Joined risk panel
Synthesizer	The three recorded panels and allowed source references	Structured proposal, confidence, rationale, conditions and referenced observations	Human narrative verification and readiness routing
Orchestrator Agent	Shared state and explicit user commands	Ordered dependencies, fan-out/join where enabled, validation, budgets and conditional stops	Next permitted node or checkpoint
Credit Memo / Report Generator	Recorded upstream outputs and readiness	Deterministic Markdown report with evidence and audit references	Existing review screens and download

The report generator is a Python service, not an additional model call. Similarly, policy validation is part of the Compliance responsibility; it is not a duplicated policy LLM agent.

New uploads use a real three-branch LangGraph subgraph after fact verification. Branches return disjoint results and independently recompute inexpensive inputs. The parent joins and persists them, avoiding concurrent SQLite mutations. Synthesis does not start until the three panels are available.

SQLite-backed LangGraph checkpoints support pause/resume. Human evidence review uses interrupts and explicit continuation. Missing facts, invalid structured outputs or unresolved provider outcomes stop progression without substituting saved answers. Local checkpoints and audit records are retained, but are not claimed to be immutable or tamper-proof storage.

8. Critical design decisions and trade-offs

Problem and alternatives	Decision and rationale	Trade-off and mitigation
One large LLM prompt versus bounded specialists	Separate extraction, Python analysis, synthesis and routing so results can be inspected independently	More state/contracts to maintain; strict schemas and scoped artifacts limit error propagation
Model-generated ratios versus ordinary code	Calculate ratios and thresholds with Decimal	Less flexibility for unusual financial definitions; missing/unsupported bases are disclosed rather than guessed
Free-running supervisor versus a stateful graph	Use explicit LangGraph nodes, dependencies and human interrupts	Less autonomous planning; predictable progression suits the narrow demo

Problem and alternatives	Decision and rationale	Trade-off and mitigation
Sequential analysis versus parallel specialists	New uploads fan out Financial/Compliance/Risk and join; sample/historical behavior is preserved	Small repeated calculations; parent-only writes avoid state races. No measured speedup is claimed
Generate a memo immediately versus show intermediate review	Keep Workspace, source navigation and checkpoint reviews	More reviewer interactions; users can identify an incorrect period or unsupported claim before relying on the memo
Large retrieval infrastructure versus bounded document context	Use current deal's eligible cited pages directly for the MVP	Limited document size; benchmark retrieval separately before changing the runtime path
Treat AI Approve as final versus maintain separate decisions	Preserve AI proposal, deterministic readiness and human decision separately	More statuses; consistent labels explain why a positive proposal may still be Pending Review
Add many formats versus finish a narrow supported path	Keep the verified runtime intake to text PDFs	CSV, XLSX, DOCX and OCR remain unsupported in this intake until parser integration is validated

These choices are documented in the project specifications and implementation. The claim that specialist separation improves maintainability at larger scale is **[DESIGN RATIONALE — INFERRED]**, not a measured scalability result.

9. Technology stack and implementation

Verified technology	Role and fit	Limitation / trade-off
Python 3.12	Application, calculations, policies and report services	Local runtime and dependency setup required
Streamlit	Six-screen Python review interface	Rerun/session behavior needs careful identity handling
LangGraph	Stateful coordination, conditional routing and specialist subgraph	More explicit graph/state code than a linear script
langchain-core	Selected framework primitives	Does not imply every agent uses an autonomous LangChain executor
Nebius Token Factory	Live generation with Qwen/Qwen3-30B-A3B-Instruct-2507	Requires credentials, network access, token fit and available allowance
Pydantic	Strict extraction, synthesis and action contracts	Valid JSON does not prove factual correctness
pypdf	Local text-PDF parsing and page references	No OCR or guarantee of complex table/year-column fidelity
Decimal	Exact arithmetic and deterministic comparisons	Does not verify the economic meaning of an input
SQLite / LangGraph SQLite checkpointer	Scoped records, retained sources, checkpoints and action history	Local persistence; no enterprise access isolation or tamper-proof audit claim

Verified technology	Role and fit	Limitation / trade-off
pytest / Streamlit AppTest	Financial, graph, authorization and UI regressions	Mocked provider tests do not establish live model quality
Optional RAGAS 0.4.3 evaluation code	Separate answer-quality evaluation	Completed saved-answer evidence must not be presented as current live-answer evaluation

Core boundaries are implemented under `src/deal_or_no_deal_ai/`: upload intake and profiles; live context/worker; graph; finance/policy/risk services; readiness; uploaded decisions; memo; and shared UI projections. Prompts and specialist contracts are versioned under `docs/agents/`; model configuration is under `config/`. API credentials remain local and must not be committed or shown.

Retrieval clarification: runtime upload analysis uses bounded, cited document context. Parsing and placing pages in a prompt is not a vector RAG pipeline. Separate benchmark code implements DIRECT, keyword, vector and hybrid retrieval mechanics; hosted embedding capability has been checked. Those components do not establish a deployed hybrid retrieval path for every uploaded review. No Pinecone or other hosted vector database is required by the current workflow.

10. Financial risk and compliance methodology

The following definitions are implemented in Python. Inputs must refer to the same borrower, applicable period, currency and units. Definitions and thresholds are educational choices, not universal lending standards.

Metric / rule	Definition and required evidence	Standard Deal result / configured check
EBIT	Net income + interest expense + income-tax expense; annual schedule	USD900,000
EBITDA	EBIT + depreciation + amortization; no discretionary addbacks	USD1,100,000
Funded debt	Existing funded debt + proposed borrowing − documented repayment	USD3,300,000
DSCR	Annual cash available for debt service ÷ (annual principal + annual interest)	900,000 / 600,000 = 1.50x; sample exception below 1.25x
Debt / EBITDA	Funded debt ÷ annual EBITDA	3.00x; sample exception above 4.00x
LTV	(Secured exposure + prior-ranking liens) ÷ eligible equipment value × 100	60.00%; sample exception above 80% when equipment policy applies
Revenue growth	(Current revenue − comparative revenue) ÷ comparative revenue × 100	10.00%, comparing supplied projections
Current ratio	Current assets ÷ current liabilities	2.00x
Interest coverage	EBIT ÷ annual interest expense	9.00x
Debt-to-equity	Requires an approved equity-based definition and inputs	**Not implemented**; Debt / EBITDA must not be relabeled as debt-to-equity

Metric / rule	Definition and required evidence	Standard Deal result / configured check
Repayment capacity	Annual CFADS and documented annual debt service	Annual coverage only; balloon/refinancing/full-term adequacy not established
Covenant readiness	Applicability, timing, due date, evidence and compliance status	Annual financial-statement obligation remains incomplete without supported evaluation

Financial schedules supply the income statement, annual debt service and pro-forma balance sheet. Collateral pages supply value, exposure and lien premises. The loan agreement supplies terms and the annual-reporting covenant. Configured sample policy also includes term, amount and confidence-based review controls. Threshold breaches identify exceptions; they do not automatically mean Reject.

The illustrative composite score weights repayment capacity 30%, leverage 20%, collateral 20%, credit profile 20%, and business/industry 10%. It aggregates authored factor ratings, not a trained default model. The Standard Deal's supplied ratings produce **21/100, Low**; this is not a 21% default probability. New generic uploads do not inherit these authored ratings and remain unscored when factors are unavailable.

Facts, arithmetic, policy findings, AI interpretations, missing evidence and human decisions remain distinguishable. Presence of a citation is not proof that its claim is supported. Future reporting obligations are not automatically breaches.

Educational approval exception: completed uploaded reviews with exclusively PENDING blockers may receive an explicitly acknowledged, reasoned human approval override from a simulated NORMAL Authorized Approver. The AI readiness remains Pending Review and the unresolved requirements remain recorded. Missing/stale evidence, material policy exceptions, recovery errors and outstanding information requests are not covered. This demonstration exception is not a real lending authorization or a model-generated waiver.

11. End-to-end demo walkthrough

Case: Example Equipment LLC, fictional USD1.2 million equipment term loan, 48-month term, supplied 2027 annual projections and 2026 comparative projections.

Demo step	User action	Agent / backend behavior	Evidence and result
Upload	Select the agreement, financials and collateral/assessment PDFs	Retain and parse one package	Original files, page text, validation and citation handles
Context	Review prefilled borrower, periods and applicability; confirm and save	Save an explicitly unverified reporting profile	Source suggestions and one upload-derived Deal ID
Start	Click Run once	Live Extraction Agent emits structured facts	Uploaded values and page references, not saved responses
Verify	Complete required checks in the indicated reviewer roles	LangGraph pauses/resumes for new upload evidence review	Recorded support or withholding, never automatic Pass
Analyze	Continue	Financial, Compliance and Risk branches join	DSCR, LTV, policy results, missing factors and risk flags

Demo step	User action	Agent / backend behavior	Evidence and result
Synthesize	Continue after applicable checks	Live synthesis proposes a referenced explanation	Original AI proposal plus separate effective readiness
Review	Inspect Workspace and Agent Activity	Read the same deal's stored artifacts	Financial inputs, rule traces, source explorer and handoffs
Memo	Open Credit Memo and download	Python assembles upstream outputs	Findings, proposed conditions, limits and supporting sources
Decide	Authorized simulated reviewer records an action and comments	Separate version-checked human event	Decision History with UTC timestamp and retained analysis snapshot

The memorable result is **Low illustrative risk, zero numerical policy exceptions, but Pending Review**. The annual-statements covenant still needs its contractual timing and requirements evaluated. A Low score cannot complete that review.

Historical uploaded Standard Deal runs completed under the earlier unverified demo-draft policy. The new uploaded path adds mandatory fact and narrative checkpoints; do not present a historical draft as proof those newer human steps were performed. A missing-data case pauses or produces an explicitly partial review rather than inventing values or displaying a fabricated low score.

12. Evaluation and testing

The following are recorded results from different checkpoints, not a single fresh full-suite run. Counts overlap and must not be added together.

Evidence	Observed result	What it establishes
Actual browser-uploaded Standard Deal	Completed, two generation requests; 4,018 input and 1,109 output tokens; generation cost USD0.0007345 at configured rates	A bounded real upload/live-analysis/source-navigation/memo-download journey
Upload acceptance regression	57 tests and 55 subtests passed	Sample compatibility, deterministic results, citations, permissions and navigation under offline mocks
General-upload module	11 tests passed, including inherited compatibility cases	Different borrower/values, parallel fan-out, checkpoint/resume, missing score and same-ID screens
Reporting-context save fix	Six focused tests passed	Metadata saving does not clear or depend on an unrelated execution lock
Educational approval override	Four focused tests passed; additional UI check passed separately	Explicit authorized override, preserved readiness, ordinary approval restrictions and visible controls
Recovery regressions	Ten tests passed	Scoped isolation retains unknown costs/history and blocks replay

An independently read completed live Standard Deal record also contained the expected nine calculated values and Pending Review readiness. No new model call was made while preparing this report.

Existing evaluation, with limits: recorded native RAGAS scores for a saved Standard Deal answer are Faithfulness 0.8, Response Relevancy approximately 0.5363 and Context Recall 0.8. Context Precision is Not Applicable for that unranked DIRECT context. The separate custom five-dimension rubric recorded 5/5 per dimension. A fact-level judge accepted 31 citation checks and flagged five of 36 targets. These are different methods and populations; none proves live uploaded answer accuracy or substitutes for human evidence review. Systematic live-answer RAGAS and broader optimization remain deferred to Phase 2.

Proposed next measurements: extraction precision/recall against authored facts; supported-claim and citation-navigation rates; risk recall together with false positive counts; unsupported-answer behavior; completion/failure rates; checkpoint correctness; stage/end-to-end latency; and actual token/cost totals. Establish explicit references and a baseline before setting release targets. No measured analyst time saving, general accuracy percentage or parallel latency benefit is claimed.

One operational lesson was directly observed: the earlier sandboxed runtime could not reach Nebius, while an approved read-only check outside it returned HTTP200 and confirmed the selected model. A network-capable local app was subsequently started. Credentials, network access and generation correctness are separate checks. Unknown request costs remain reserved rather than inferred from rounded aggregate bills.

13. Business value and differentiation

The demonstrated value is a connected review record: documents become cited facts, facts become reproducible calculations, calculations feed explicit rules, and findings become a reviewable memo and separate human action history.

Potential institutional benefits include reduced transcription, consistent application of approved rules, quicker draft preparation and clearer ownership of unresolved issues. These remain hypotheses for a future user study, not validated productivity or financial-return claims.

Differentiation is visible in concrete behavior: a two-call generation design avoids spending tokens on arithmetic; current-period/source scope prevents unrelated-deal leakage; partial information is exposed; and original AI proposals are retained even when effective readiness disagrees. Those are useful product and engineering demonstrations without claiming enterprise readiness.

14. MVP limitations and prioritized roadmap

Priority	Next work / limitation	Value, complexity and risk
P0	Finish one newly uploaded general-profile live walkthrough with genuine human evidence review	High demo value; moderate effort; closes the gap between offline generalization tests and live use
P0	Stabilize one documented app launch and improve provider diagnostics	High usability/reliability value; low-to-moderate effort; prevents confusing old-server and unknown-request states
P0	Reconcile historical uncertain provider usage when matching records are available	Cost transparency; external evidence dependency; no invented usage
P1	Phase 2 live-answer evaluation on held-out synthetic cases	High trust value; moderate effort; measure unsupported claims and excessive risk flags

Priority	Next work / limitation	Value, complexity and risk
P1	Test unfamiliar financial layouts, year-column alignment and cut-off passages	High error-reduction value; moderate effort; maintain explicit unsupported states
P1	Benchmark DIRECT, KEYWORD, VECTOR and HYBRID on identical questions	Determine retrieval value before adding complexity; do not assume vector search is better
P2	Additional formats, OCR and larger document packs	**[FUTURE ENHANCEMENT]** High breadth, higher parsing/context risk
P2	Real identity/authorization, institutional policy governance and protected audit infrastructure	**[FUTURE ENHANCEMENT]** Required before considering real financial-service deployment

Current intake is bounded to up to three synthetic text PDFs, ten pages and 10MB, USD whole units and annual analysis. Separate statistical default prediction, calibrated PD/LGD/EAD, real bureau/KYC integrations, portfolio concentration validation, multi-user load testing and production security are not established. The notebook artifact does not have a verified fresh Jupyter-kernel run; optional notebook installation was deferred. No public deployment is claimed.

15. Deliverables and links

Deliverable	Submission status
Application and specialized graph	Implemented locally; see architecture and limitations above
Sample memo and live evidence	Retained locally; select an approved synthetic artifact for submission
Specification-driven documents	PRD, HLD, feature specs, execution plan, evaluation and traceability files
GitHub	**[INSERT GITHUB LINK — NOT VERIFIED]**
Demo recording	**[INSERT VIDEO LINK — NOT VERIFIED]**
Public application	**[NOT DEPLOYED]** Local instance is not a shareable hosting link

Suggested recording: show one package, inspect a source page, run the graph, explain a financial calculation, open its supporting evidence, demonstrate why Pending Review differs from Reject, and download the same deal's memo. Clearly label any educational human override and any saved response used in a fallback presentation; do not represent saved output as a new live call.

16. Conclusion

Deal or No Deal AI demonstrates a financial review workflow whose reasoning is inspectable at each handoff. Its strongest example is not an automatic approval: it is the ability to explain why apparently healthy financial metrics still leave an unresolved contractual review. The combination of bounded specialization, deterministic controls, source navigation and separately recorded human judgment makes the project a useful educational multi-agent implementation.

Evidence index

- Product requirements: PRD.
- Architecture: HLD, uploaded fan-out decision.
- Upload behavior: general upload contract, human decisions.
- Actual live/browser evidence: uploaded live results.
- Functional regression: acceptance fixes.
- Earlier evaluator evidence: answer-quality results.
- Runtime diagnosis: network context.
- Code: `graph.py`, `live_worker.py`, `finance.py`, `policy.py`, `risk.py`, `readiness.py`, `uploaded_decisions.py`, `memo.py` under `src/deal_or_no_deal_ai/`.

Some referenced documents retain historical status sections. Current code and explicitly dated execution evidence take precedence over an older status sentence.